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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING™

Management of Business Administration-Technology Management (Information Technology)

Financial Management

Topic: Business Plan of Rooftop Solar Panels

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About Business: Rooftop Solar

Rooftop solar panel installation refers to the setting up of solar panels on residential or commercial buildings for the purpose of tapping solar energy and converting it into electricity. This kind of renewable energy has several advantages such as lower power bills, environmental friendliness, and self-reliance in terms of power. Some parts that make up such a system are **Photovoltaic**



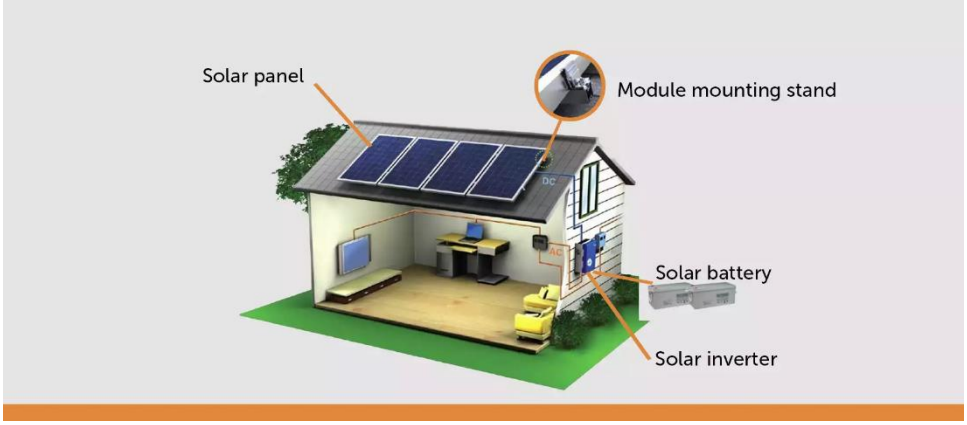


modules, Mounting systems, Cables, Solar inverters, and other electrical accessories.

What is Rooftop Solar? Rooftop solar is a photo voltaic system that has its electricity-generating solar panels mounted on the rooftop of a residential or commercial building or structure. Rooftop mounted systems have capacities in the kilowatt to megawatt range. Most rooftop PV stations are Grid-connected photovoltaic power systems.

Solar Panels produce carbon free electricity when irradiated with sunlight. Often made of Silicon, solar panels are made of smaller solar cells which typically have 6 cells per panel. Multiple solar panels strung together make up a solar array. Solar panels are protected by tempered glass and secured with an aluminium frame.

How does a Solar Panel work? Solar rooftop panels rely on photovoltaic cells within the panels to absorb the energy emitted by the sun and convert it into electricity. These solar cells are made of silicon, receive sunlight, a chemical reaction occurs within them, they release electrons, which generates an electric current. The core component of these photovoltaic cells in a solar rooftop system is the layers of doped silicon crystal-based semiconductor materials. When the cells receive sunlight, photons knock electrons out of both the layers. The flow of electrons then occurs, the electrons then travel to an external circuit through which electricity is carried for usage purposes, controlled by the P-N junction.



How is this converted electricity utilised and tracked? The electricity is in the form of DC current, that is unusable for home appliances. A solar inverter converts it into AC current, flows through a service panel, after which it can be safely used for all purposes. Simultaneously, a solar meter keeps track of the solar rooftop power production and brings to notice any potential problems. This is the basic process through which solar energy is converted into useable electricity.

What we do?

1. **Consultation and Site Assessment:** We start with the evaluation of the feasibility of the solar installation site. This is done by assessing such factors as roof positioning, exposure to shadow, extent of structural stability, typical local climates, and amount of sun available.
2. **System Design:** After undertaking a site assessment and obtaining knowledge on client's power needs, they create a tailor-made photovoltaic system. These involve identifying potential spots for solar panels, choosing correct types of panels and inverters, and developing electrical layout.
3. **Permitting & Paperwork:** For one to install solar on their premises, there are different permits that ought to be sought from local authorities. We provide that service as well.
4. **Procurement of Equipment:** We purchase all required solar panels along with their components like inverters, mounting hardware etc., which are considered essential for installation purposes. We have direct connections with suppliers or manufacturers who always provide quality equipment.
5. **Installation:** The main service that the Company provides. Then we send experienced professionals who will install the solar panels, mount them securely on the roof and connect to electrical grid or battery storage if there, the installation process follows safety measures with an aim of averting accidents and making sure that the system operates as required.

Why this Business?

- **Energy Independence:** Rooftop solar installations provide some form of energy independence and reduce the dependence on regular fossil fuel-based power sources, lessening the effects of power outages.
- **Rising Electricity Demand:** The high demand for electricity seen nowhere before brings up solar power as a green solution to this problem diminishing reliance on fossil fuels and ensuring energy security.
- **Favourable Government Policies:** Solar energy aspirations are evidenced by ambitious goals and supportive policies. Roof top solar mandates, net metering plus other initiatives create an investment friendly environment for infrastructure development in India.
- **Clean and Green Energy:** Rooftop solar panels generate electricity using sunlight that is available in abundance and can be replenished. Unlike fossil fuels, solar power generation does not cause any greenhouse gas emissions or air pollution that makes it clean energy.
- **Increased Home Value:** Normally, houses with rooftop solar installations have higher valuation than those without. These panel are highly desirable to environmentally conscious buyers and tenants.
- **Grid Stability and Reliability**: By relieving stress on centralized power systems, distributed rooftop solar installations help improve grid stability as well as reliability of the same. This way, solar energy eases congestion in transmission and distribution networks by producing electricity at proximity to its end users.
- **Scalability and Modularity:** Rooftop solar installations can be adjusted to suit the specific energy requirement of every house, business, or community. Moreover, they are also capable of being easily constructed from small beginnings so that they can accommodate changing requirements for energy in future.

Pain Points of the Business

- **High initial investment:** This hampers installation of these panels on rooftops. Securing the necessary capital can be challenging particularly to small enterprises or startups. Many homeowners and businesses steer away because of the cost involved during the beginning.
- **Difficult processes:** The knowledge of how to install solar panels on roofs may discourage some prospects from considering green energy sources. Fear about building permits, structural soundness and maintenance are insurmountable obstacles.
- **Unawareness and inadequate information:** Most customers have a limited understanding of how rooftop solar panel installations work as well as about the benefits of solar energy. It is

important therefore that the market be educated regarding solar power advantages and its practicality.

- **Grid Integration:** Integrating large-scale solar into the existing grid necessitates significant infrastructure upgrades and the implementation of smart grid technologies to manage variable energy generation effectively.
- **Complicated permissions and regulations:** Rooftop solar installations have complex permitting systems as well as local regulations. This involve going through zoning laws, building codes and utility regulations thus increasing bureaucracy levels which require expertise in this area.
- **Competition:** There is an increase in competition among companies in the rooftop solar industry who offer almost similar services. Standing out amidst competitors can be quite hard especially for new entrants into such sector.
- **Technical Challenges:** The need for expertise in technical areas like solar technology, electrical engineering, and structural engineering both in designing and installing rooftop solar systems. Well-trained personnel who is skilled in installation needs to be backed by proper training system for safety and efficiency purposes.
- **Supply Chain Issues:** Solar panels, inverters, mounting hardware among other components are mostly outsourced from suppliers. In cases where there is dependence on them, however, it can cause supply chain disruptions, delays or even quality problems.
- **Financing Options:** Barriers including upfront costs could be solved through providing financing options or lease arrangements to the customers. Nonetheless, this may not be an easy task as obtaining finance and structuring attractive financing packages to customers becomes a problem especially to small businesses.
- **Seasonality and Weather Dependency:** Demand for rooftop solar installations may vary according to the season with sunny months having peak demand. However, weather conditions can also affect installation schedules thus affecting earnings and business operations.

Elaborate Opportunity (Segment-wise)

1) Residential Segment:

- Homeowners: A growing number of homeowners are interested in rooftop solar installations. They do so to cut down their electricity costs, improve property value and show eco-friendliness. Good deals in enticing financing options, seamless installation processes, and custom-made solutions based on individual requirements can be taken advantage of.
- Residential Communities: Construction projects that involve creation of residential houses, flats or blocks also provide good business environment for bulk sales or community solar power projects. Partnerships with real estate developers, homeowners' associations (HOA) and

utility firms may help boost the widespread adoption of rooftop solar energy across communities.

2) **Commercial and Industrial (C&I) Segment:**

- Small and Medium-sized Enterprises (SMEs): Businesses within the SME sector can benefit from rooftop photo voltaic installations through cost cutting, embracing corporate social responsibility (CSR) initiatives, as well as enhancing brand value. Customized funding packages, energy audits for efficiency, and performance guarantees are some of the strategies that could encourage solar energy use by SMEs.
- Large Corporations: Many big companies have targets on carbon reduction as well as sustainability hence demanding for rooftop solar to be one of the clean energy goals. The appropriate solutions in this area involve provision of power purchase agreements (PPAs), turnkey solutions with other innovative financial arrangements that address their unique needs.
- Industrial Facilities: There are big roof tops in manufacturing industries, warehouses and other industrial facilities which can be utilized for solar power generation. In addition, it is possible to integrate solar with building infrastructures or minimize energy consumption depending on its usage for each facility thereby taking advantage of economies of scale for cheaper deployment costs.

3) **Public Sector:**

- Government Buildings: There is a good opportunity to install solar power systems on the roofs of federal, state, and local government buildings. Public agencies are often more interested in ecological issues and saving money. Such institutions can be considered for installation of solar plants because they often collaborate with governmental organizations, ESCOs as well as financial institutions that help develop projects.
- Educational Institutions: Roof-top solar is being used in schools, colleges, and universities as a means of cutting down on operational costs, promoting green initiatives and providing learning opportunities for students. Available options range from pay-performance contracts to educational alliances through to capitalizing on incentives/grants.

Competitor Analysis

There are Public and Private Sector Companies as Competitors:

Sectors	**(PSUs) Public Sector:**	**Private Sector:**
Companies	NTPC, THDC, SVJN, NHPC, Power Grid Corporation and etc	Tata Power, Reliance Energy, Adani Renewable Energy, Vikram Solar, Waaree Solar etc

Financial Support	Dedicated Support of Central Government.	Dedicated support of international companies.
Market Dominant	No	Yes
Working Area's	Mostly in Govt Missons and for supplying to high energy projects like Hydrogen energy.	Mostly to the Customers requirements (Commercial and Residential).

Long standing solar installation companies: Some of the established players in the rooftop solar panel installation sector may include well-known names in the industry. These firms are likely to have a significant market presence and brand recognition. For Example: Citizen Solar, Tata Power, Vikram Solar, Waaree Solar etc.

Small scale suppliers: Small-scale contractors or smaller entities could also offer rooftop solar installation services. For instance, while they may lack resources compared to large-sized firms, such entities could compete on some aspects including individualized attention and prices.

Emerging businesses and Startups: The increasing popularity of solar energy has seen many that have just entered or yet to enter business taking part in it too. The operations of these companies may be concentrated around advanced technology adoption, unique means of finance or even focusing on market segments.

Total Financing required

This section outlines the amount of capital needed to expand the business. It should include:

- Total Investment: 3,00,00,000.00/- (2cr Debt and 1Cr @10% Equity).
- Breakdown of Costs: The Amount will be used for expanding in the new locations. To be clearer we want to Expand our Business in cities like Jaipur, Chandigarh, and Lucknow in FY25 and In FY26 we want to enter Noida, Gurugram also in the Private commercial and industrial (C&I) and public sector in the existing cities. The amount will be used to create the Offices in the cities and warehouses.
- Use of Funds: 60,00,000.00/- will be used to create Offices and warehouses in the 3 new cities i.e., Jaipur, Chandigarh, and Lucknow FY25 to start operations and 80,00,000.00/- will be used to create Offices and warehouses in the 2 new cities i.e., to start operations in Noida and Gurugram. Also, when we will expand to Private commercial and industrial (C&I) and public sector in we need high capital to counter already existing companies like Tata Power, Reliance Energy, Adani Renewable Energy, Vikram Solar, Waaree Solar, etc.

Some Questions Answered

1. **What problem does your product solve?**
   Ans. Rooftop solar generates electricity, it also has multiple uses, such as "Regular house use, heating water and providing heated air for the building or institution. Moreover, India being a tropical country with most of the day being lit with sunlight, rooftop solar has become a convenient renewable energy source.

2. **Tell us why your product is unique. What are the distinguishing characteristics?**
   Ans. Solar rooftops need only the light of the sun to generate electricity, making it a cleaner source of energy than most other usual forms. Its renewable nature promises sustainability. No health hazards are involved, and no pollutants are emitted by solar rooftops. No extra land is required to set up rooftops.

3. **What are your sales figures and growth projections?**
   Ans. As of FY24 our cost of (installation+ GST) price is 1,84,750.00/- and the selling price is 2,75,000.00/- which makes our Gross Margin for Installing 5Kw is 32.82% and we have installed 104 units on an average making it to the 520Kw and revenue is 2.75Cr and making the PAT approx. 51 Lakhs. And we project to Install 1000Kw in the FY25
   These figures should show whether the business has been profitable, what its margins are, and how it expects to grow in future.

4. **Indicate how you will acquire customers.**
   Ans. We have started in Tier 2 city (Bhopal), i.e., the city has huge amount of customer that can install Rooftop Solar for their homes to reduce electricity bills. And we do not compromise the quality of the product we install to our customer, currently the quality of made in India Solar panel is not at world class level hence we import our solar panel from the China which is manufacturing hub of solar panel.

5. **Tell me how you make money?**
   Ans. As of FY24 our average cost of (installation+ GST) price is 1,84,750.00/- and the average selling price is 2,75,000.00/- which makes our Gross Margin for Installing 5Kw is 32.82%. We have improved our margins over the years,

6. **How much funds are you raising and what equity are you offering for investment?**
   Ans. We are Offering a 10% of the Equity for 1 Cr and additionally we are also raising debt of 2Cr at 19%, without collateral.

7. **What are your capacities regarding delivery?**
   Ans. We hire the Labour (costs us 0.07% of SP) for the installing the Plant and at least one of the permanent employees is present at the site of installation to check everything is completed with precision.

8. **What are the Problems that your current business model faces?**
   Ans.Problems we face in the current Business model is that Indian want, made in India Product but we must import the Panels from China to maintain quality of our product installed.

9. **How do you intend to utilize this investment cash?**
   Ans. Currently we are only in the residential segment, and we want to expand to the Private commercial and industrial (C&I) and public sector as 72 per cent and 11 per cent of the total market, respectively in commercial and industrial (C&I) and public sector, while the residential market's share was estimated at 17 per cent.

10. **Why Should you Invest?**
    Ans. Moving ahead with the investment we intend to acquire larger share of the smaller customer segment and expand our operations into multiple cities. We are expecting good returns on the investment and ROA. We are expecting an IRR of 90% and healthy cashflows in the business. We intend to grow but profitably and not by burning cash in expansion. We are also working on improving margins and compete with big players in town. In the coming years we will also share a part of the profits earned by the company and try make good on your investment.